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Human–AI Interaction in Healthcare: A User-Centered Evaluation of Trust and Usability in Explainable AI Interfaces

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Abstract & Introduction

- AI-based healthcare systems often suffer from low adoption due to lack of transparency, which reduces user trust and confidence.
- This study evaluates how Explainable AI (XAI) affects trust, usability, and confidence in a healthcare decision-support interface.
- A diabetes risk prediction prototype was developed with two interface types: black-box (prediction only) and explainable (prediction with feature-based explanations).
- Results from a controlled user study show the explainable interface significantly improves user trust, perceived usability, and confidence compared to the black-box interface.
- The findings highlight the importance of human-centered explainable interface design for building trustworthy and widely accepted AI systems in healthcare.

Objectives

- To design and develop a machine learning–based healthcare decision support system for diabetes risk prediction.
- To implement two interface types—black-box and explainable—to compare the role of transparency in AI systems.
- To evaluate the impact of explainability on user trust, usability, and confidence in healthcare AI interfaces.
- To conduct a controlled user study using standardized metrics (e.g., trust score and SUS) for quantitative comparison.
- To derive design insights and recommendations for building trustworthy, user-centered explainable AI systems in healthcare.

Methodology

- Perform a focused literature review on Explainable AI (XAI), Human-Computer Interaction (HCI), and AI in healthcare to identify research gaps.
- Collect and preprocess healthcare dataset(s), including cleaning, normalization, handling missing values, and feature engineering.

Methodology

- Design the proposed HCI-based XAI framework integrating predictive model(s) with explainability techniques and user-centered interface.
- Train and evaluate the model using appropriate metrics (e.g., accuracy, F1-score, AUC) and assess explanation quality and usability through user-interaction analysis.
- Conduct comparative experiments with baseline models and analyze performance, interpretability, and user trust to validate the effectiveness of the system.

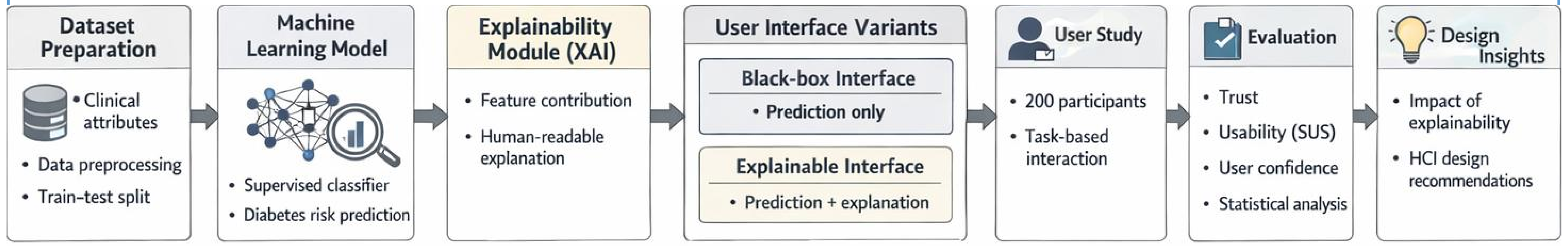
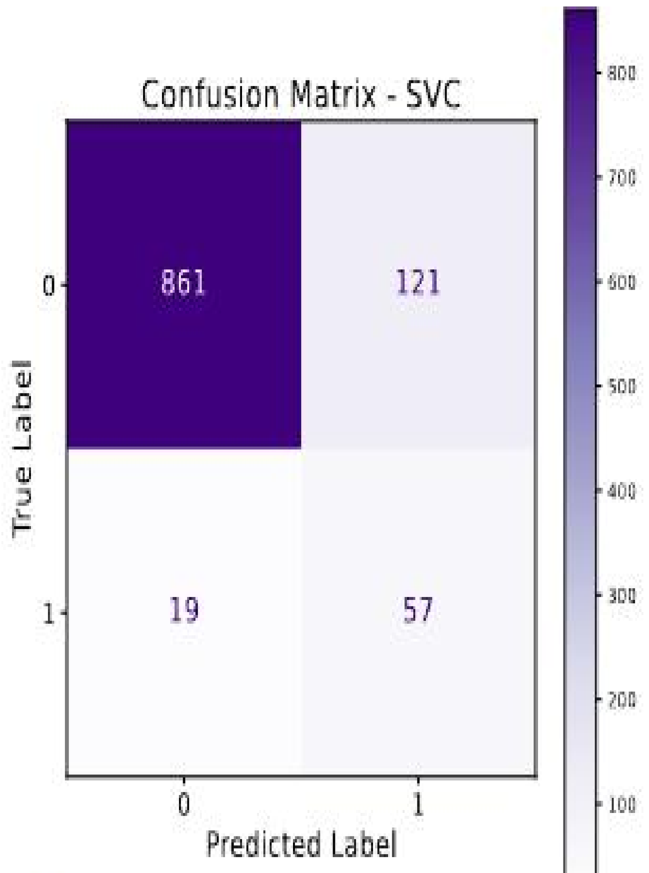
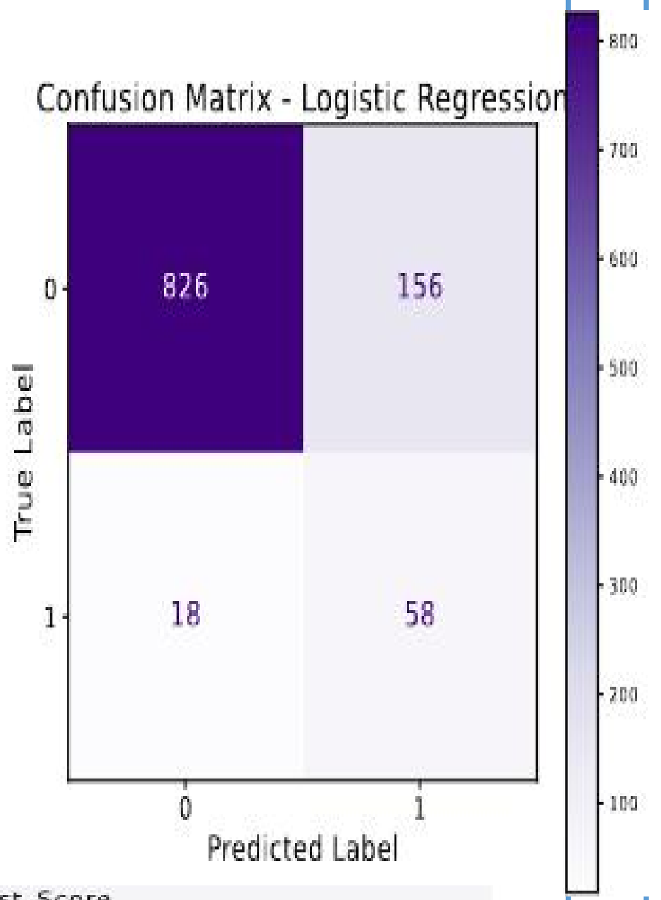


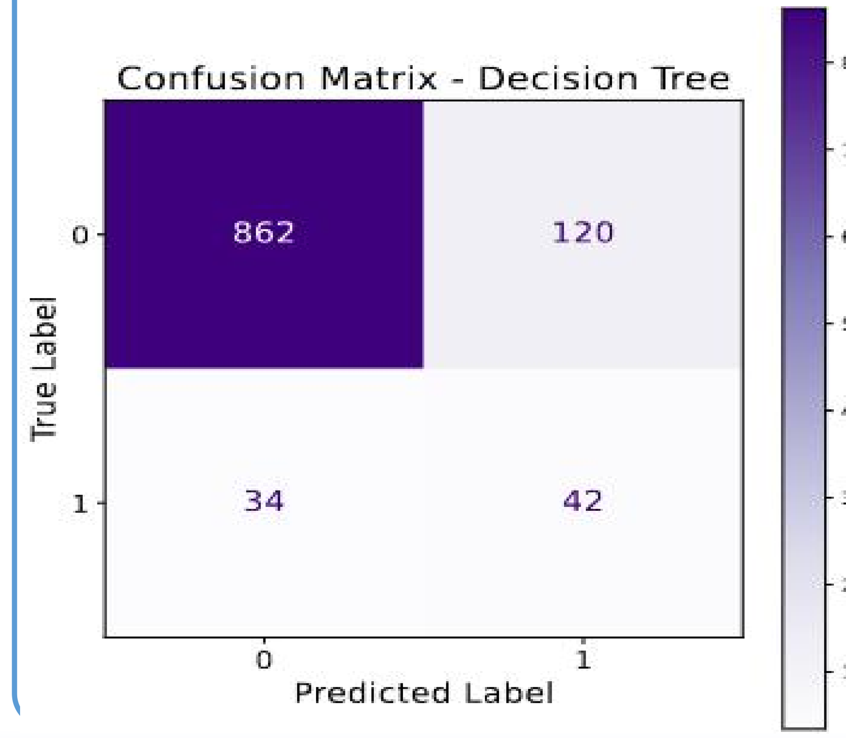
Figure 1. Workflow of the Proposed HCI-Based Explainable AI Healthcare System

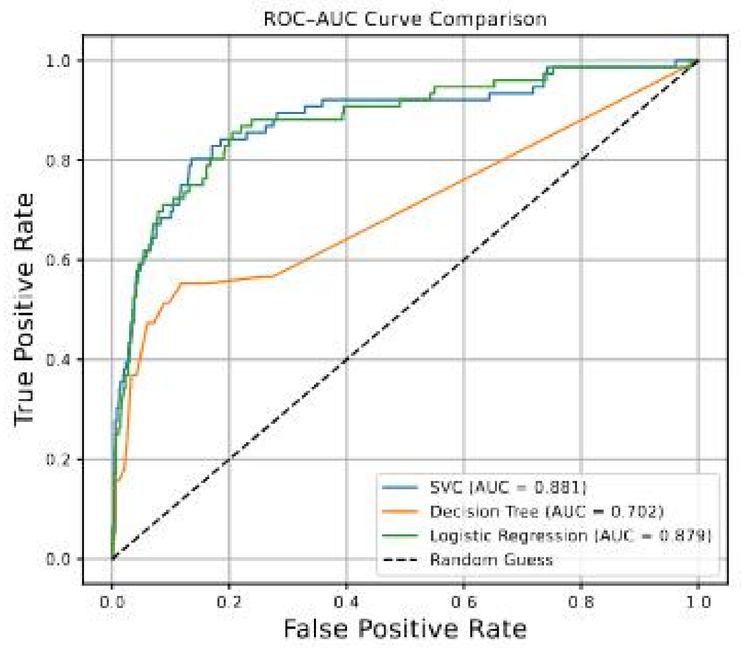
Result

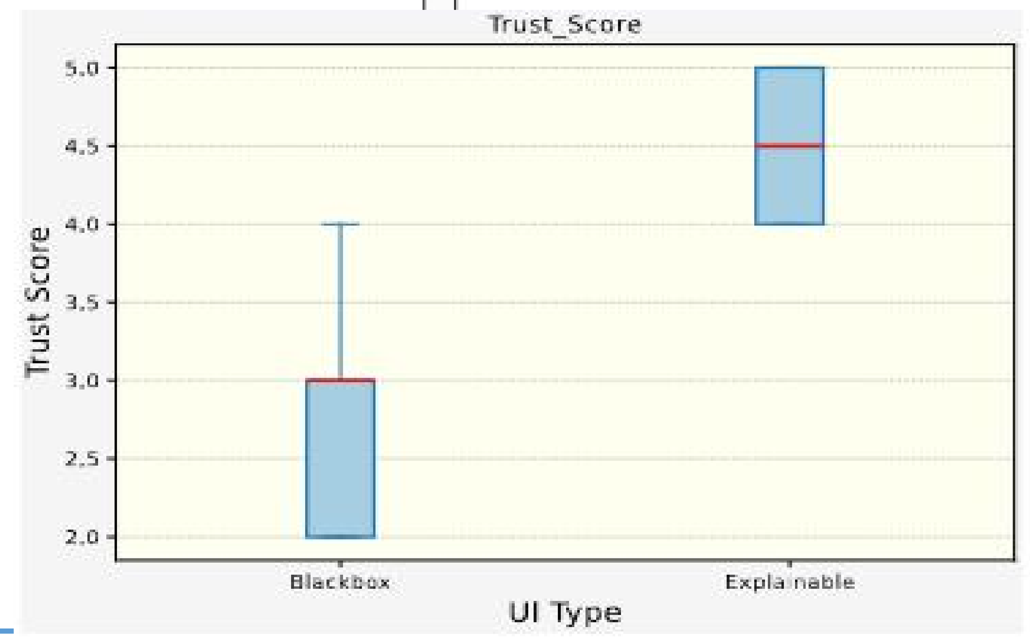
Algorithm	Accuracy (%)	Precision (%)	Recall(%)	F1-Score (%)
SVM	93.57	93.35	92.46	92.54
Logistic Regression	93.47	93.23	92.34	92.29
Decision Tree	90.59	90.85	90.52	90.56



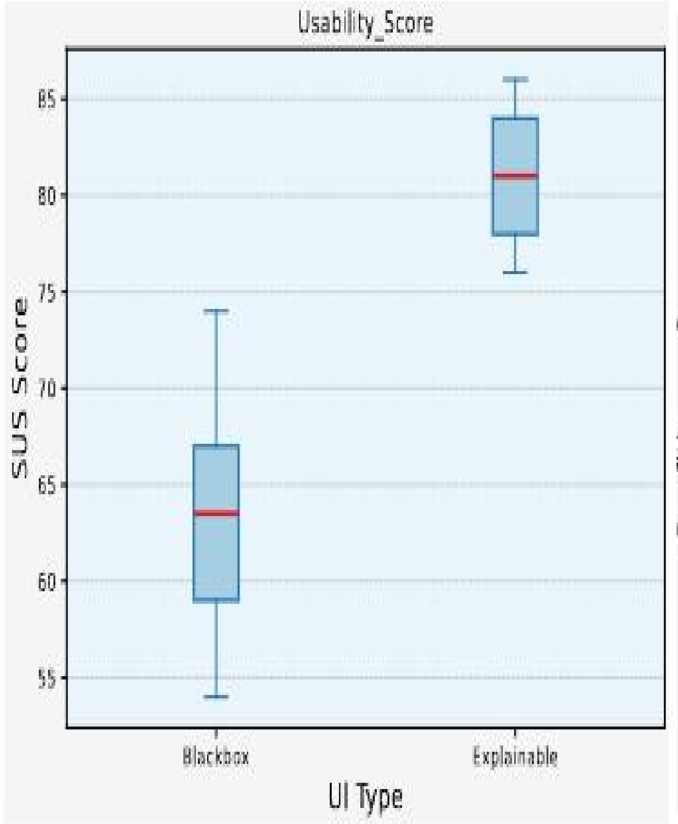


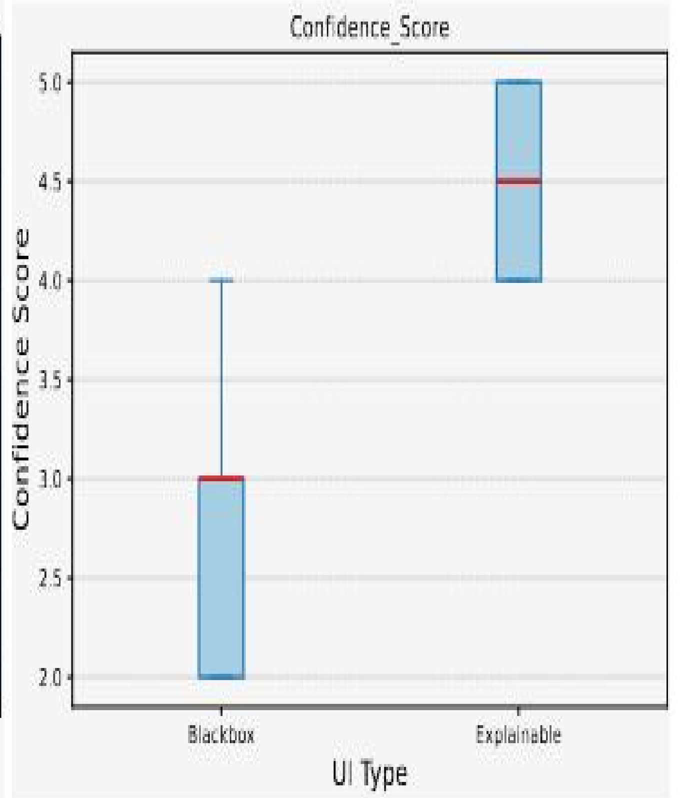






Result





Conclusion & Future Work

- The proposed HCI-based Explainable AI system improves prediction and provides clear explanations.
- Results show that explainability and user-centered design improve decision support and trust.
- Human-in-the-loop evaluation helps measure usability, interpretability, and user trust.
- Future work will use larger real-world clinical datasets for better generalization.
- Advanced explainability, adaptive interfaces, and longitudinal studies will support real-world adoption.

References

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